Module 5: Installing and configuring Hyper-V and virtual machines

ACIT 3420

Johnny Zhang

Agenda of Hyper-V

- What is Hyper-V?
- New Hyper-V host features in Windows Server 2022/2019
- Installing Hyper-V
- Configuring networking on Hyper-V host servers
- Configuring Hyper-V virtual machines
- Managing virtual machines

Overview Virtual Machine

- By using ***virtual machine*** technology, you can run multiple operating systems concurrently on a single machine, which allows separation of services while keeping cost to a minimum.
- It can also be used to create Windows test systems in a safe, self-contained environment.
- Microsoft Hyper-V is a hypervisor-based virtualization system for x64 computers starting with Windows Server 2008.
- The ***hypervisor*** is installed between the hardware and the operating system and is the main component that manages the virtual computers.

What is Hyper-V?

- Hyper-V is the hardware virtualization role in Windows Server 2022/2019
- The hypervisor controls access to hardware
- Hardware drivers are installed in the host operating system
- Many guest operating systems are supported:
  - Windows Server 2008 SP2 or newer
  - Windows Vista SP2 or newer
  - Linux
  - FreeBSD

New Hyper-V host features in Windows Server 2022

New features for Hyper-V hosts include:

- Host resource protection (prevents the excessive use of CPU Usage)
  (Set-VMProcessor -EnableHostResourceProtection $True)
- Hyper-V Manager improvements (Manage earlier versions..)
- Nested virtualization (Educational Purpose)
- Rolling Hyper-V cluster upgrades
- Shielded virtual machines
- Start order priority
- Storage QoS
- Windows PowerShell Direct

New Hyper-V virtual machine features in Windows Server 2022

New features for virtual machines include:

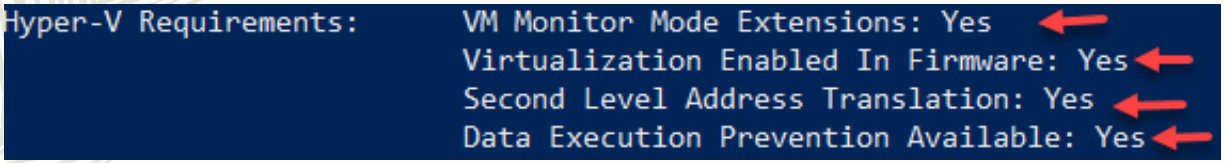
- Discrete device assignment (DDA) (Peripheral Component Interconnect Express PCIE)
- Hot add or remove for network adapters and memory
- Integration services delivered through Windows Update
- Key storage Drive(Trusted Platform Module (TPM))
- Linux Secure Boot: verifies digital signatures on files during the boot process to prevent malware.
- Memory and processor capacity improvements (12 TB, 240 Virtual Processor.)
- Standard checkpoints (Standard Checkpoints (Application-consistant) vs Production Checkpoints (Data-consistant)) vs. (Backup)
- Virtual machine configuration file format (.XML vs. .VMCX, .VMRS)
- Virtual machine configuration version

New Hyper-V virtual machine features in Windows Server 2019

- Easier two-node clusters (Router+USB+Share & Permission (Witness share to server as the quorum))
- Improved Storage Spaces Direct (S2D) Performance and Limits (1Pb->4PB;)
- ReFS deduplication added
- Windows Admin Center – improved S2D visibility
  - Historic view of performance and health for S2D
  - Centralized management plane for Hyper-V Clusters
- Smaller Server Core container image （5GB->1.58GB）
- Shielded VM improvements (Linux VMs are now supported for shielded VM protection.)

Prerequisites and Requirements for Installing Hyper-V

- Use Systeminfo.exe to verify that hardware requirements are met for Hyper-V



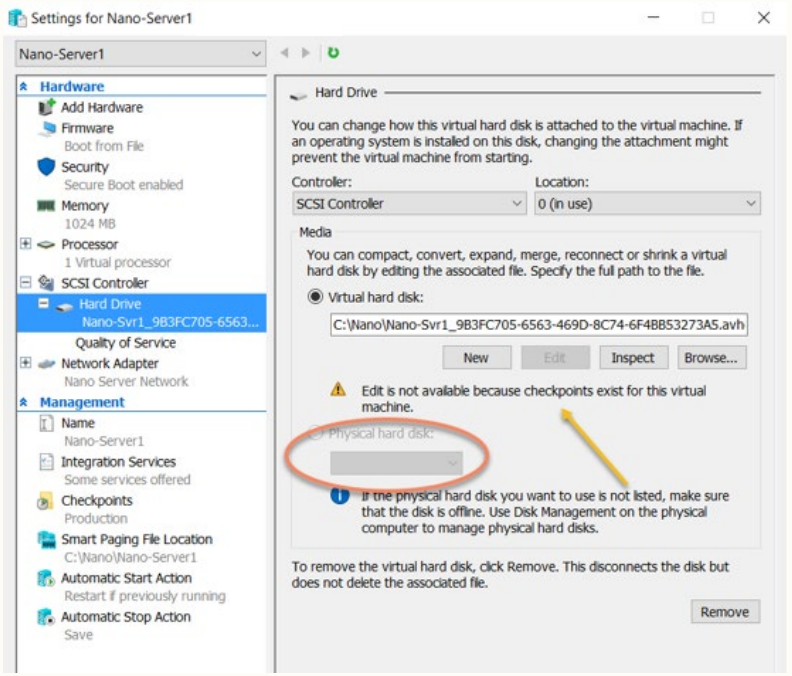
- You must have sufficient resources in the host to meet the requirements of the virtual machines:
  - Processor
  - Memory (8GB/Test)
  - Storage (SAN/Enterprise, S2D/Test)
  - Network(IGB/ Test, 10GB/Enterprise)

Nested virtualization

- Enables a Hyper-V guest virtual machine to also be a Hyper-V host
- Useful for development and test servers
- Requirements:
  - At least 4 GB of static memory
  - Windows Server 2022 or Windows 10 host operating system
  - The Hyper-V host and guest virtual machines running Hyper-V must be the same build
- Some features are not available in the guest virtual machine running Hyper-V

Storage Options in Hyper-V

- Virtual hard disk formats:
  - ✓.vhd
  - ✓.vhdx
  - ✓.vhds This is the VHD Set file, which is what your virtual machines will connect to.
  - ✓.avhdx: This is the advanced VHDX file or differencing/checkpoint disk, which is where all of your data is actually stored.
- Virtual hard disk types:
  - ✓Fixed-size (production environment)
  - ✓Dynamically expanding (test environment)
  - ✓Pass-through (Do not use it)
  - ✓Differencing



Considerations for Virtual Hard Disk Formats and Types

- Create .vhdx virtual hard disks unless you need backward compatibility with Windows Server 2008 or Windows Server 2008 R2
- A dynamically expanding .vhdx-formatted virtual hard disk is suitable for production workloads
- The free space shown by dynamically expanding virtual hard disks is not equal to physical free space
- Multiple layers of differencing disks decreases performance
- If you modify a parent disk, the differencing disk is no longer valid
- You can relink a differencing disk to a parent disk

Types of Hyper-V Networks

- Use Virtual Switch Manager to create different types of virtual networks:
  - External
  - Internal
  - Private
- You can also:
  - Configure VLANs
  - Capture data travelling through a switch
  - Filter data travelling through a switch

What are virtual machine configuration versions?

- Configuration versions allow compatibility for virtual machines between:
  - ✓ Windows Server 2012 R2 (version 5 bios?)
  - ✓ Windows Server 2022 (version 5 or 6 UEFI?) (8 above)
  - ✓ Windows Server 2019 (Version 5 or 6 UEFI) (9 above)
- Virtual machines must be manually updated from version 5 to version 6:
  - ✓ Update-VMVersion “VMName”

Virtual Machine Generation Versions

Generation 2 virtual machines provide the following functionality:

- Secure boot
- Boot from a virtual hard disk that is connected to a virtual SCSI controller
- Boot from a virtual DVD that is connected to a virtual SCSI controller
- PXE boot by using a standard Hyper-V network adapter
- UEFI firmware support

The Hot Adding Feature in Hyper-V

- Hot adding requires Generation 2 virtual machines
- Memory:
  - Add memory to virtual machines with static memory while they are running
- Network adapter:
  - Add or remove network adapters to virtual machines while they are running

Shielded Virtual Machines

- A shielded virtual machine is:
  - ✓Protected from anyone with access to the Hyper-V host
  - ✓A Generation 2 virtual machine with a virtual TPM
  - ✓Protected by BitLocker
- The Host Guardian Service has two attestation modes:
  - ✓Admin-trusted attestation
  - ✓TPM-trusted attestation

Virtual Machine Settings

- Integration services allow virtual machines to access Hyper-V services
- Smart paging allows disk to be used temporarily during virtual machine startup, when it is highly needed
- Resource metering monitors resource usage of virtual machines for planning
- Discrete device assignment allows direct access to PCIe devices
- Linux and FreeBSD Generation 2 virtual machines can use secure boot

Best Practices for Configuring Virtual Machines

- Use Dynamic Memory unless an application does not support it
- Avoid using differencing disks
- Configure multiple synthetic network adapters
- Store each virtual machine's files on a separate volume

Managing Virtual Machine State

Virtual machine states define what resources are being used:

- Off
- Starting
- Running
- Paused
- Saved

Managing Checkpoints

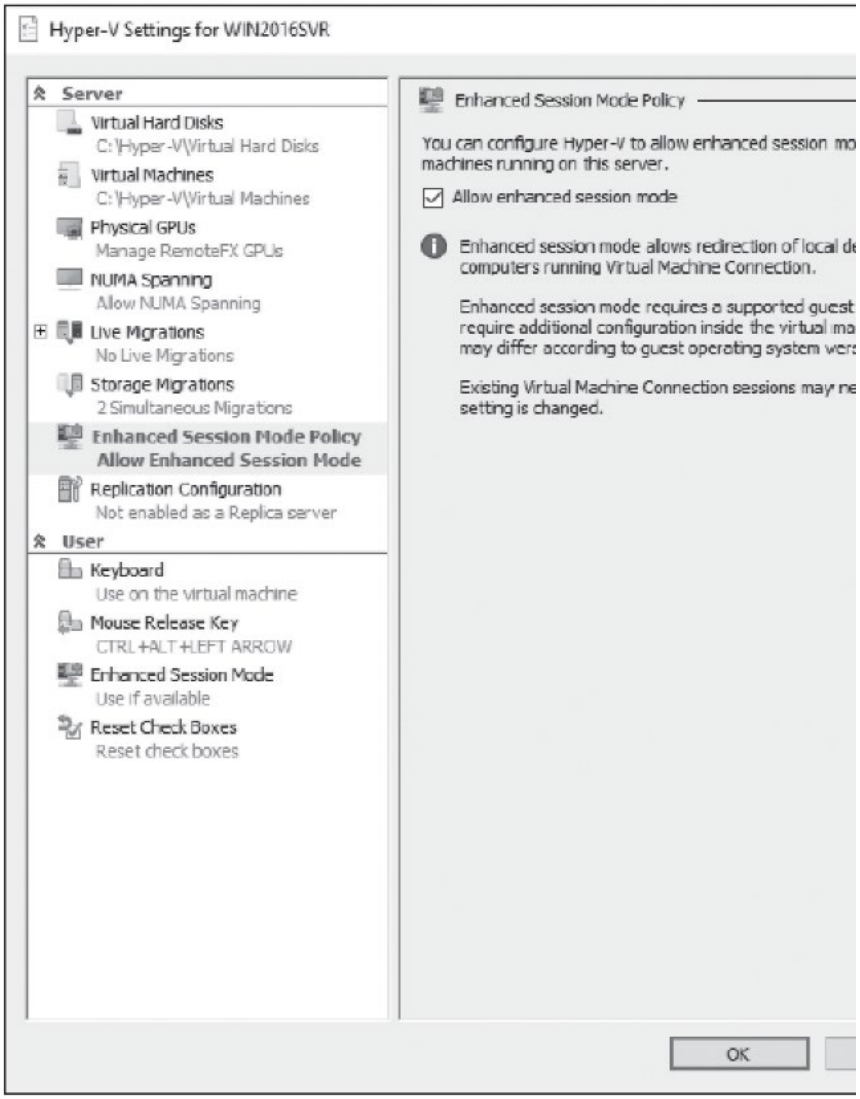
- Checkpoints allow administrators to make a snapshot of a virtual machine at a particular point in time
- Checkpoints do not replace backups
- Standard checkpoints create differencing disks, .avhd files, which merge back into the previous checkpoint when the checkpoint is deleted
- Production checkpoints are created by using VSS and require starting from an offline state

Importing and Exporting Virtual Machines

- Import options:
  - ✓Register the virtual machine in-place (use the existing unique ID)
  - ✓Restore the virtual machine (use the existing unique ID)
  - ✓Copy the virtual machine (create a new unique ID)
- Export options:
  - ✓Export a checkpoint
  - ✓Export a virtual machine including checkpoints
- Moving virtual machine storage:
  - ✓Move all the virtual machine's data to a single location
  - ✓Move the virtual machine's data to different locations
  - ✓Move the virtual machine's virtual hard disks

Windows PowerShell Direct

- PowerShell Direct:
  - Does not require network connectivity
  - Can only be used from the host to the virtual machine
- Requirements:
  - The host must be running Windows Server 2022 above or Windows 10
  - The guest must be running Windows Server 2022 above or Windows 10
  - Windows PowerShell must be running as administrator
  - You must use credentials to authenticate to the virtual machine
- Enter a session or invoke a command:
  - Enter-PSSession –VMName *VM1*
  - Invoke-Command –VMName *VM1* –Scriptblock {*commands}*



Implementing Enhanced Session Mode

- Starting with Windows Server 2012 R2, Hyper-V supports enhanced session mode.
- By using ***enhanced session mode***, you can connect to a running VM with the same features as Remote Desktop Services (RDS), even if the virtual machine does not have network connectivity or does not have the Remote Desktop feature enabled on the guest operating system.